

EC-arithmetic speedups from lopsided thin products: screening (2026-10-06)

Questions and method in the **protocol** (README.md). Rounds 1-4: thin-product interface (merged PR #1485), twelve-target survey (PR #1493), refutation audit (PR #1494), general-regime envelope (PR #1501). Status labels: **[proposed]** / **[derived]** / **[verified]** / **[measured]** (none).

Headline

No speedup: no EC kernel has the technique's shape. The method needs many wanted inner products over one shared thin middle with small-integer entries at large N. Our kernels are constant-size field formulas, sequential group operations, single inner products, or single-RHS transforms. The nearest miss dies twice over, in closed form.

Kernels vs gates; entry-size plane

No EC kernel reaches size: every block sits at shape or ring

Left: kernel funnel. Right: the (N, entry-size) plane -- our kernels sit hundreds of bits above the feasible wedge.

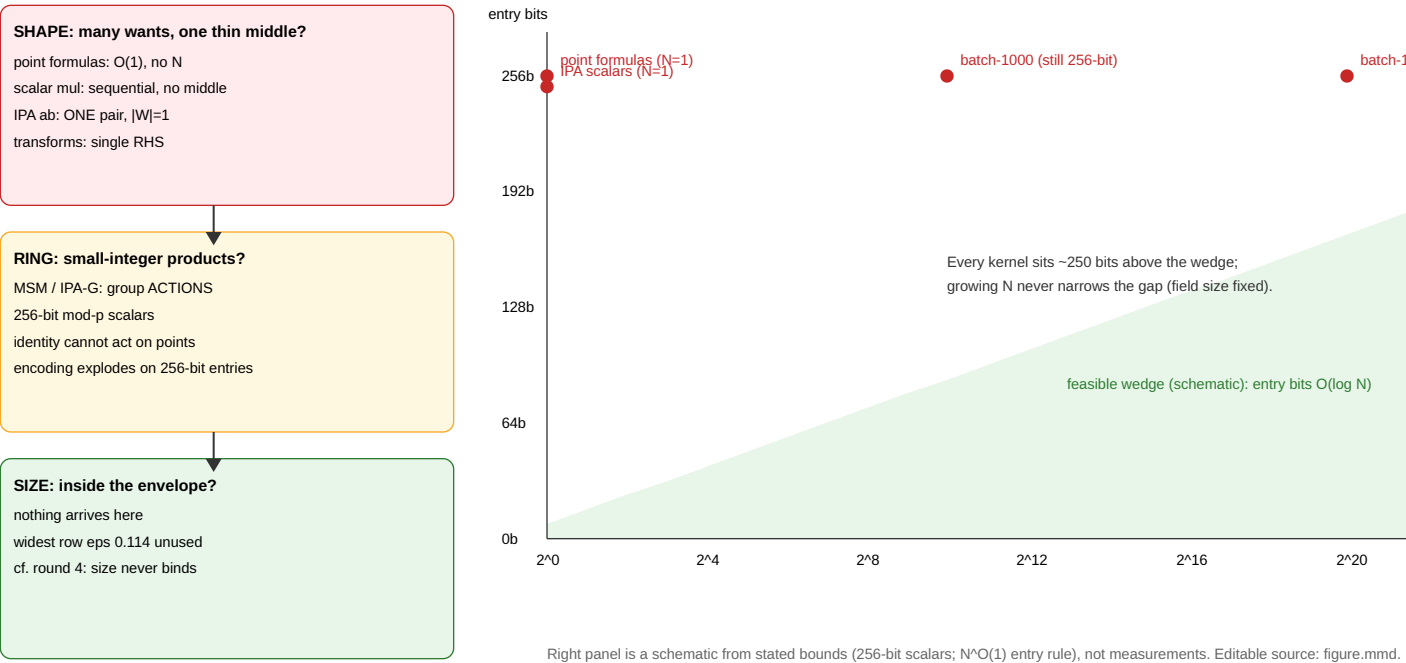


Figure 1 -- left: every inventoried kernel stops at shape or ring. Right: schematic (N, entry-bits) plane -- our kernels sit ~250 bits above the feasible wedge at every batch size. Editable source: figure.mmd.

Screening table [derived]

Kernel	Verdict	Blocking gate
Point add/double (src/ecc/point.rs)	Negative	Scale+shape: $O(1)$ field formulas, no N
Single scalar mul	Negative	Shape: sequential double-and-add, no middle
Batch verify / MSM / IPA-G / KZG	Negative	Ring: group actions, not ring products
IPA <a, b> (nearest miss)	Negative	Shape then ring: $ W =1$, 256-bit scalars
Extension fields, Miller loop	Negative	Scale+shape: $O(1)$ maps, sequential
Field multiplication; batch transforms	Negative	Scale / shape: single products, single RHS

Why the nearest miss misses

- **One-shot reading bound.** Query time is quoted after preprocessing that already read both matrices. For $|W|=1$, total work is bounded below by reading the inputs -- the naive inner product. The method's power is entirely shared preprocessing over $|W| \gg 1$; one pair amortizes nothing.
- **Entry-size bound.** Entries must be $N^{O(1)}$; ours are 256-bit field elements at every batch size. Even a hypothetical many-pair setting over F_p fails before size.

Revisit hook [proposed]

A kernel that is many small-integer inner products over one shared thin middle with known sparse wants earns a measurement protocol (matched operation counts, identical inputs/outputs), screened first against the widest `table2.json` row. No such kernel exists in the tree; until one is named, this screen stands.

Graphs checked -- no change [verified by inspection]

- Scoreboard, progress-timeline, leaderboard files, browser data, curve registry: nothing landed.
- Figure 1 is a new visual for this round, not a canonical-graph update.

Sources

- Paper: <https://arxiv.org/pdf/2610.06783> (v1); Theorems 24-25.
- Rounds 1-4 study directories (PRs #1485, #1493, #1494, #1501).
- Kernel sites: `src/ecc/point.rs`, `src/ecc/schnorr.rs`, `src/zk/bulletproofs.rs`, `src/zk/kzg.rs`, `src/bls12_381/`.